Supplementary Step 6 Visualization Report

Overview

Metric	Value
Representative geometry session	PID001 / 2025-11-06_008
Included participant-sessions from revised Step 6	35
Included questions in Step 6 trial table	30
Abstract onset-locked epochs used	524
Concrete onset-locked epochs used	521
Epoch window	-3.5 s to 13.2 s around question onset
Baseline	-2.0 s to 0.0 s
Four-ROI descriptive montage	21 pairs (42 chromophore channels)
Best raw primary ROI p-value	Anterior Ventral: 0.084645
Best Holm-corrected primary ROI p-value	Anterior Ventral: 0.338581

Adaptation note. This supplementary report was updated for the revised Step 6 rerun that uses the four ROI definitions (Anterior Dorsal, Anterior Ventral, Posterior Dorsal, Posterior Ventral) and the updated downstream exclusion policy that removes PID027. The descriptive topographies use the union of the revised Step 6 analysis ROIs rather than the earlier front/back layout.

Rendering note. As in the earlier supplementary visualization reports, the sensor-geometry figure uses the participant digitization point cloud plus source, detector, channel, and pair geometry rather than an fsaverage brain-surface rendering, because the local `pyvista`/`fsaverage` stack is not available in this environment.

Step 6 Main Figures Reused

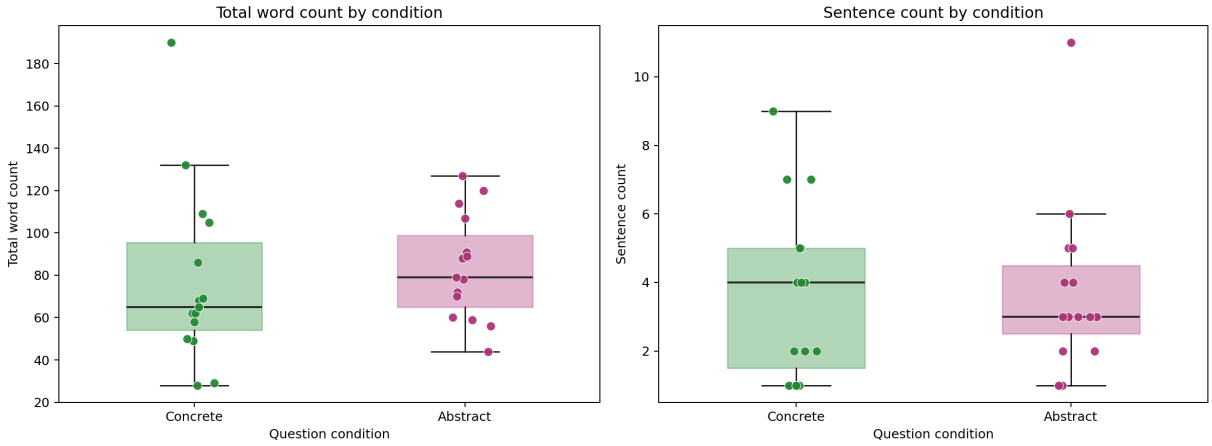

Figure 1: Step 6 item-level balance plots for total word count and sentence count.

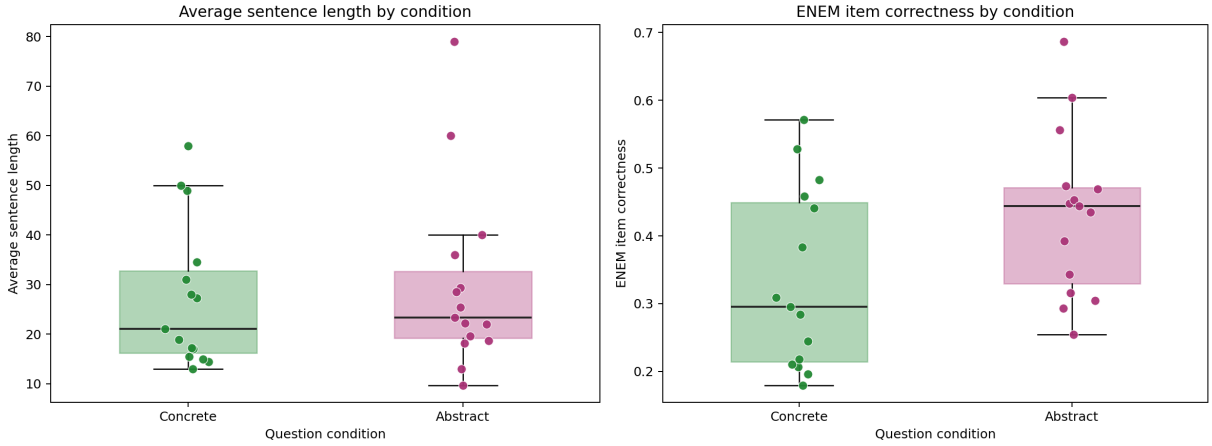

Figure 2: Step 6 item-level balance plots for sentence length and ENEM item correctness.

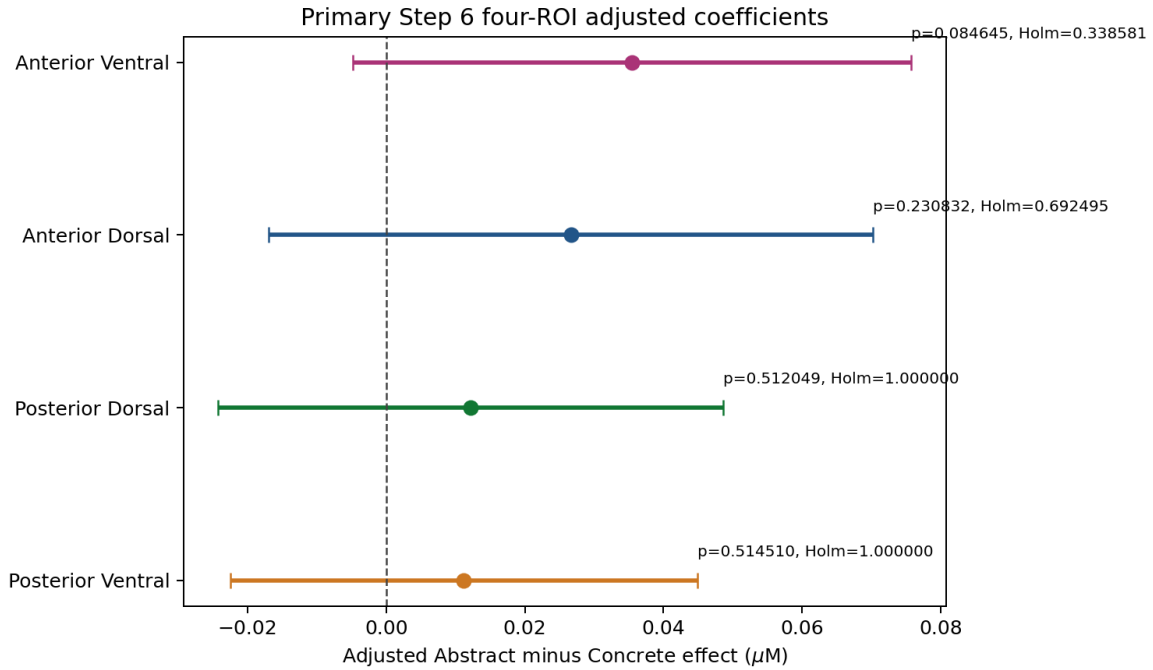

Figure 3: Primary adjusted Abstract-minus-Concrete coefficients for the four revised ROIs, with 95% confidence intervals and Holm-corrected p-values.

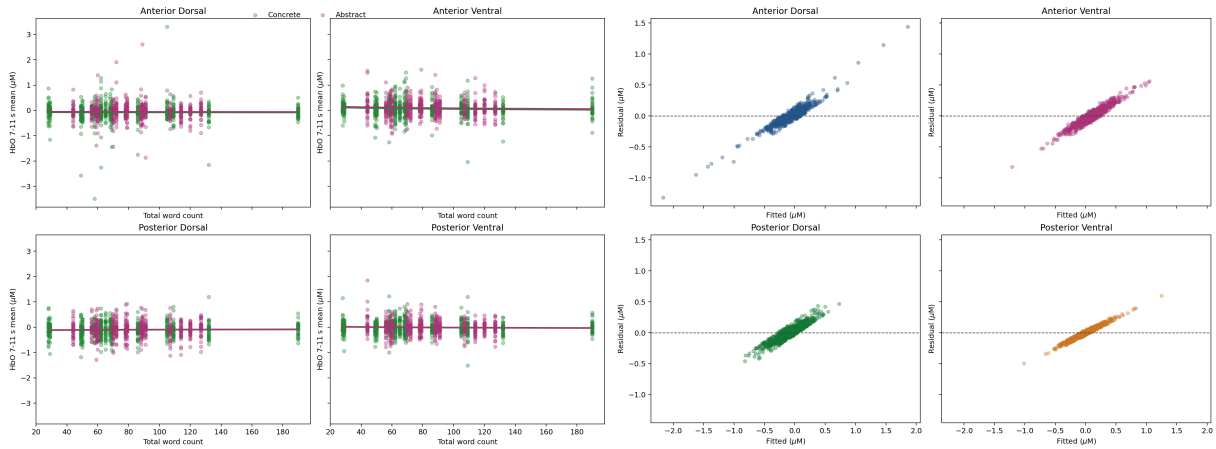

Figure 4: Primary revised Step 6 ROI-versus-word-count plots and residual diagnostics across the four ROI-specific models.

Step 5-Style Supplementary Plots for Step 6

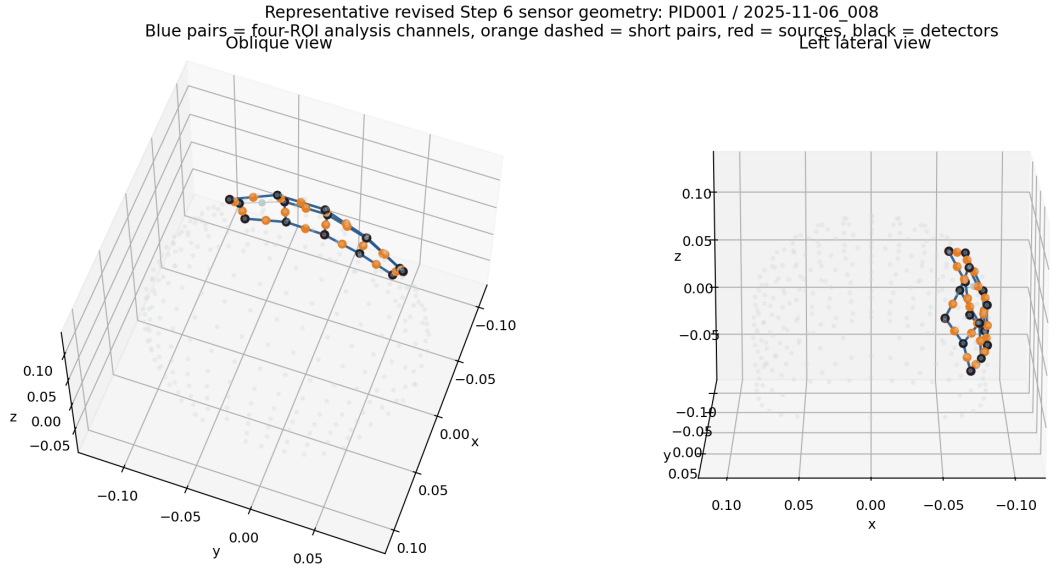

Figure 5: Representative optode, channel, and source-detector-pair geometry for one Step 6 included participant-session. Blue pairs mark the revised four-ROI analysis montage and short pairs are shown with dashed orange lines.

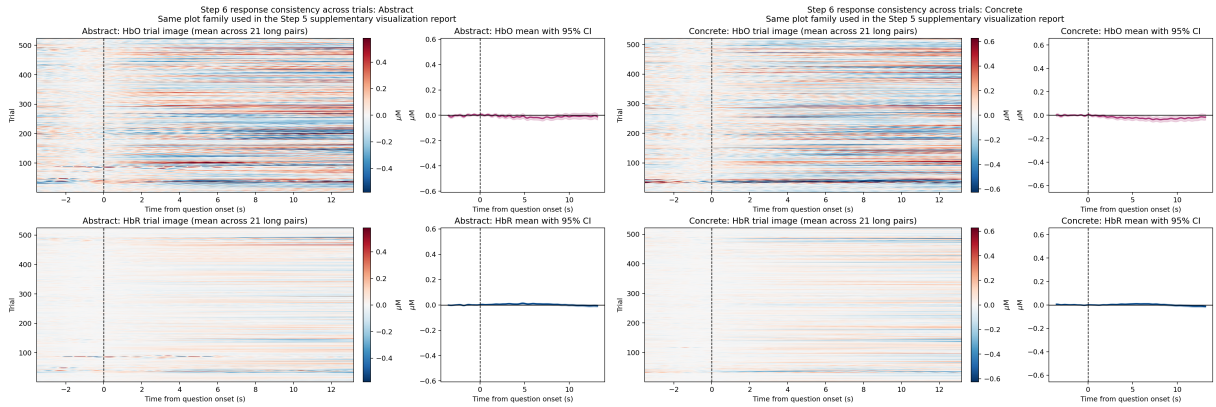

Figure 6: Condition-adapted trial-consistency image plots for the Step 6 trial set.

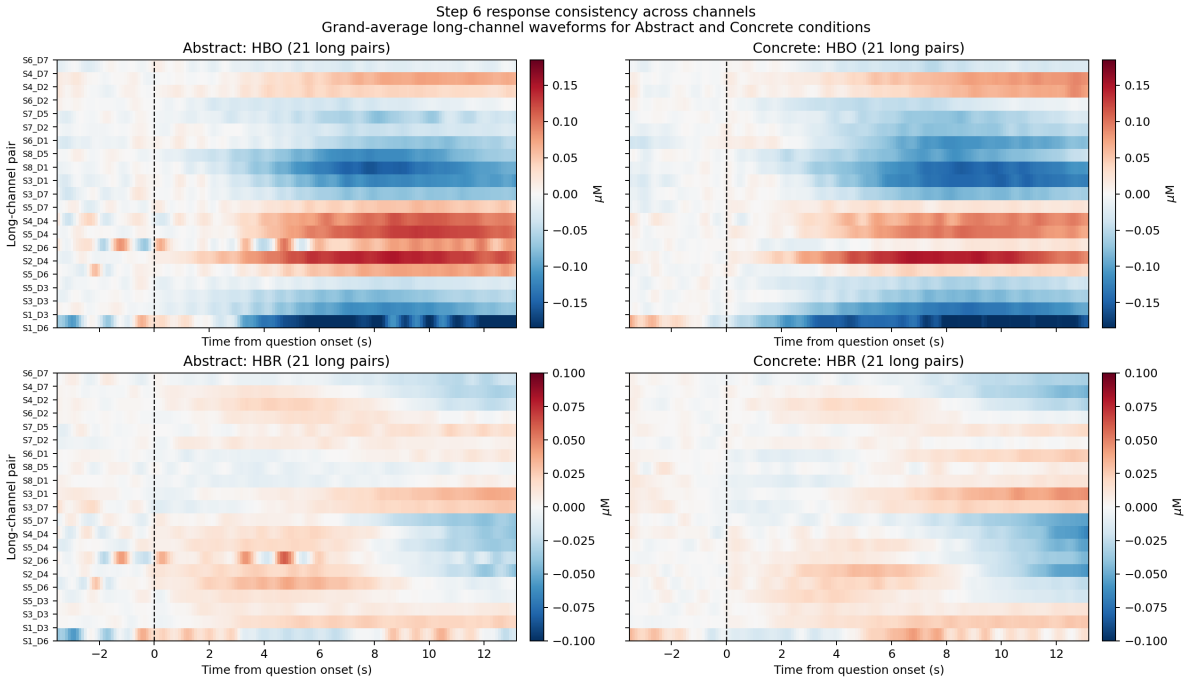

Figure 7: Grand-average long-channel response images across channels for the Abstract and Concrete conditions.

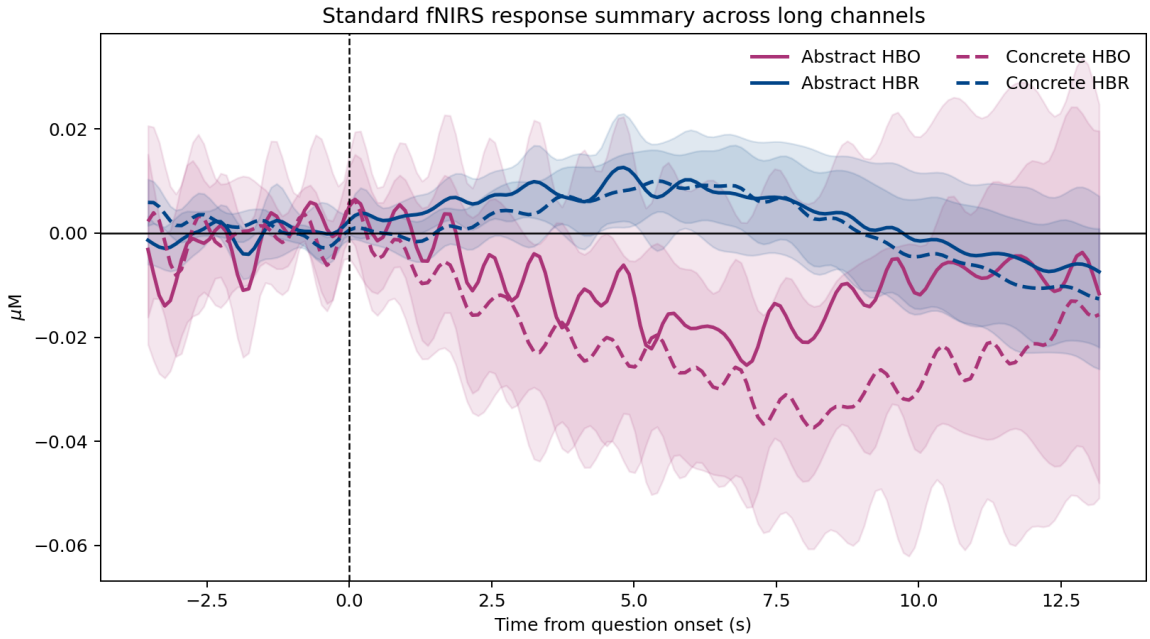

Figure 8: Standard fNIRS waveform comparison across long channels for the Step 6 trial set.

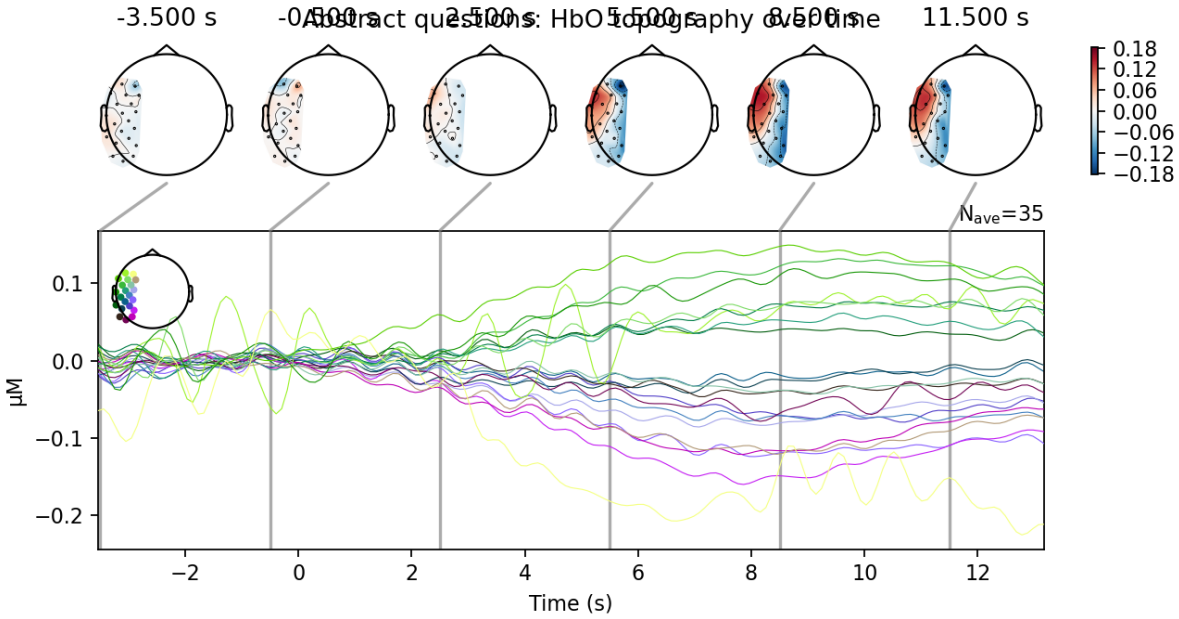

Figure 9: Grand-average HbO topography over time for Abstract questions using the revised Step 6 included trial set and restricted to the four-ROI analysis channels.

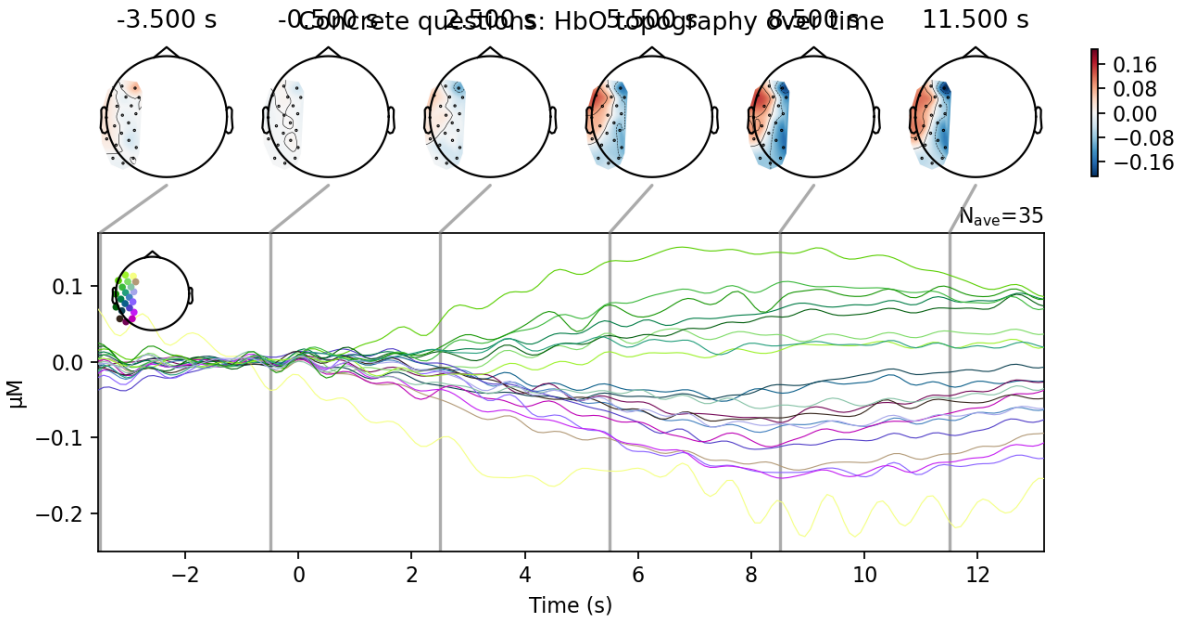

Figure 10: Grand-average HbO topography over time for Concrete questions using the revised Step 6 included trial set and restricted to the four-ROI analysis channels.

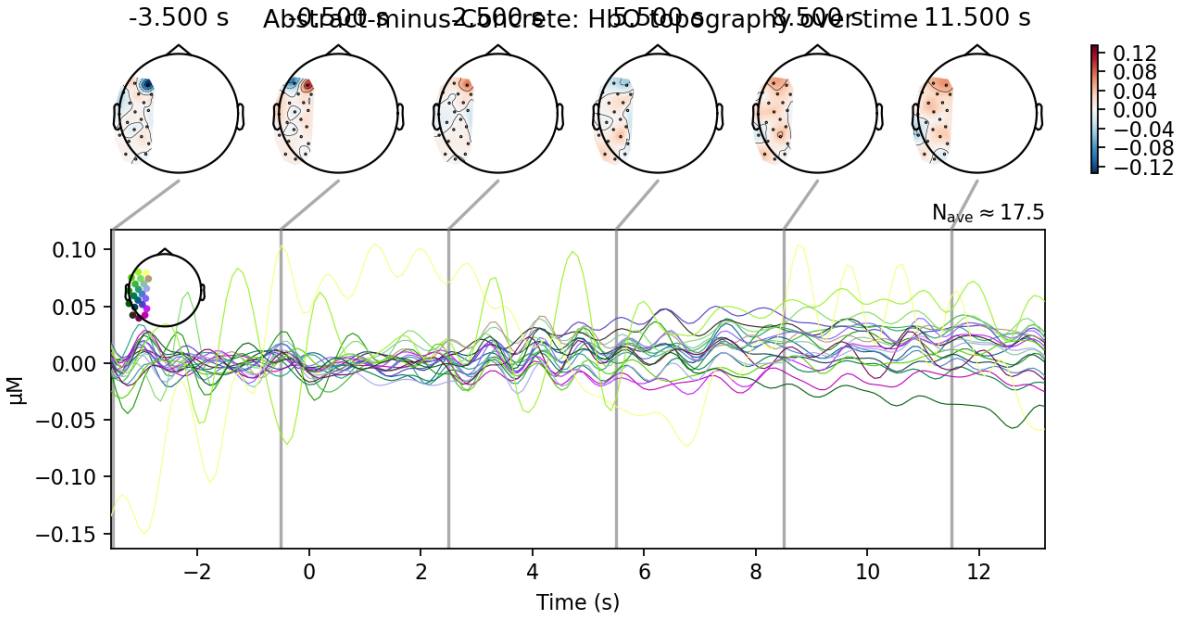

Figure 11: Grand-average Abstract-minus-Concrete HbO topography over time.

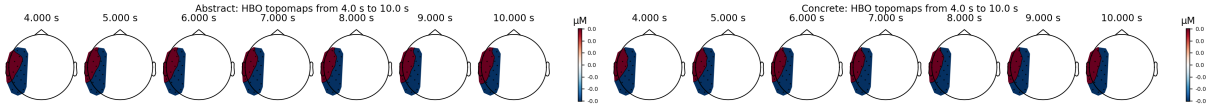

Figure 12: HbO topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

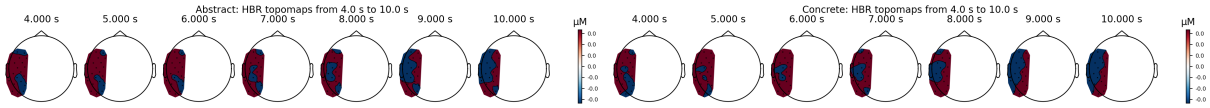

Figure 13: HbR topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

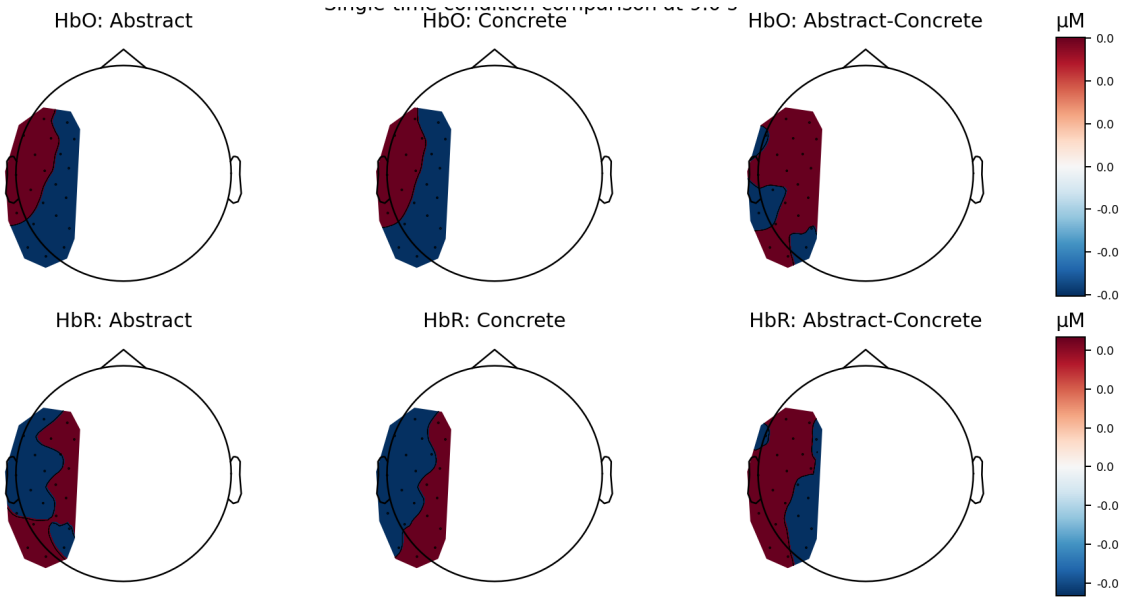

Figure 14: Single-time comparison at 9.0 s for Abstract, Concrete, and Abstract-minus-Concrete topographies for both HbO and HbR.

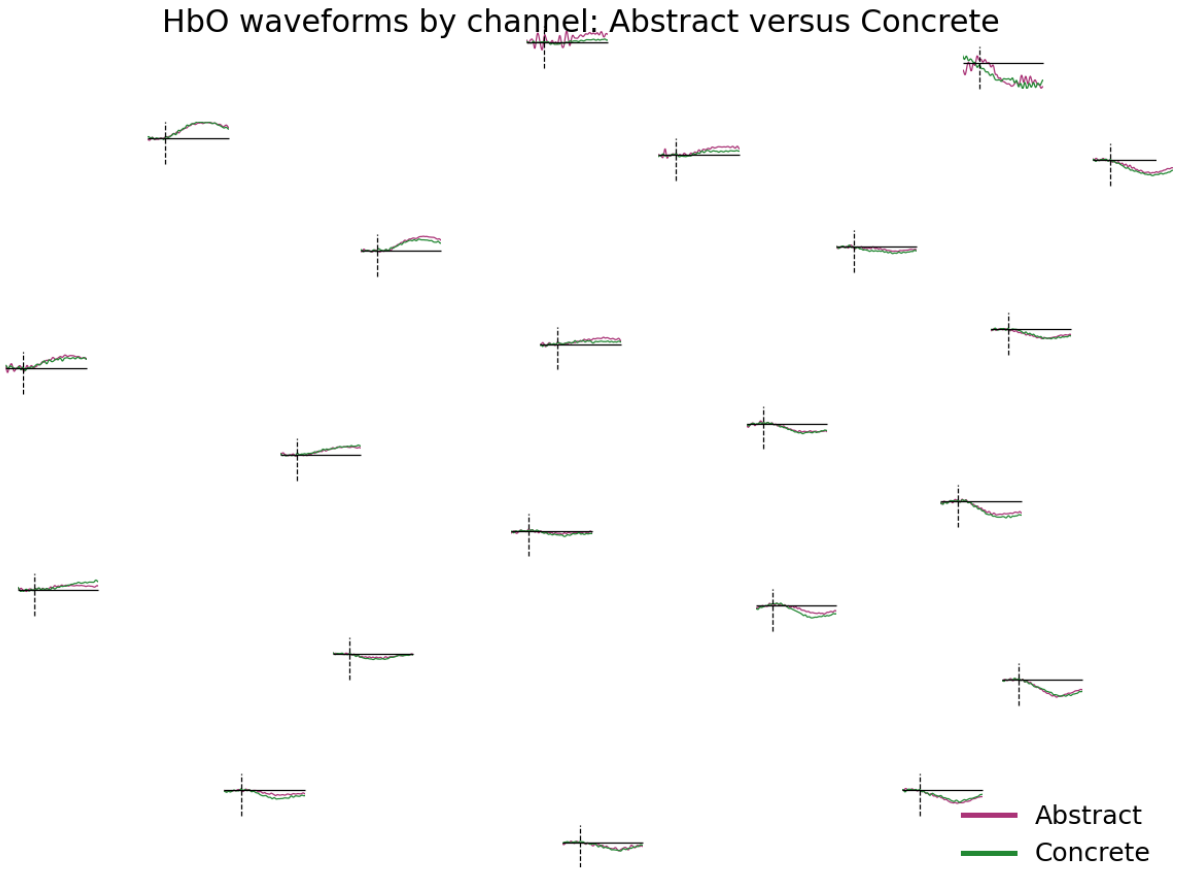

Figure 15: HbO waveforms by long channel, overlaid for Abstract and Concrete, to show what drives the descriptive topographic differences.